Udit Deo

Ph.D. Scholar in Adaptive Visual Recognition under Distribution Shift

Indian Institute of Technology Roorkee

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Profile

Ph.D. scholar at the Machine Vision and Intelligence Lab, IIT Roorkee, investigating reliable and compute-efficient visual recognition under distribution shift. My current research focuses on adaptive inference in vision transformers—specifically, how calibration degradation under domain shift undermines confidence-guided computation, and how lightweight shift-aware mechanisms can restore allocation reliability for both image classification and dense prediction tasks. Previously, my work addressed fuzzy relation inequalities and multi-objective optimization frameworks for decision-making under uncertainty.

Research Experience

2025–Present **Reliable Adaptive Computation under Distribution Shift**
Machine Vision and Intelligence Lab, IIT Roorkee

- Studying confidence-guided adaptive computation in vision transformers, examining its reliability under synthetic corruptions (ImageNet-C) and real-world adverse conditions (ACDC).
- Identifying how calibration degradation under distribution shift undermines confidence-based compute allocation, leading to unreliable token-level decisions.
- Designing lightweight, shift-aware mechanisms that combine prediction confidence with feature-space divergence from clean-data representations to restore allocation reliability.
- Evaluating adaptive token merging for image classification and extending the framework to dense prediction via SegFormer-based semantic segmentation.

Education

2025–Present **Indian Institute of Technology Roorkee**
Ph.D. in Computer Science and Engineering, GPA: 8.14/10
Advisor: Prof. R. Balasubramanian. Research area: reliable and efficient computer vision under distribution shift.

2023–2025 **Maulana Azad National Institute of Technology Bhopal**
M.Tech. in Software Engineering, GPA: 9.53/10
Secured 2nd rank in the program. Thesis: *Bridging Fuzziness and Optimization: Frameworks for Relation Inequalities and Multi-Criteria Decision Systems*.

2017–2021 **Shri Mata Vaishno Devi University**
B.Tech. in Computer Science and Engineering, GPA: 8.21/10
Received a tuition-fee waiver for securing 2nd rank in the seventh semester.

Publications

2025 **Robust Client–Server Quality Allocation via Ordered Weighted Averaging: A Multi-Objective Optimization Framework**
Udit Deo, J. K. Jain, and M. M. S. Beg
International Conference on Computing, Communication and Networking Technologies (ICCCNT).

2025 **A New Approach to Fuzzy Relation Inequalities: Effects of Variable Absence and Weighted Composition**
Udit Deo, J. K. Jain, and M. M. S. Beg
International Conference on Artificial Intelligence and its Application (ICAIA), pp. 127–138. doi:10.1007/978-981-95-0493-0_10.

2025 **Advances in Fuzzy Relation Inequalities and Optimization Techniques**
Udit Deo, J. K. Jain, and M. M. S. Beg
International Conference on Computational Intelligence, Communication Technology and Networking (CICTN), pp. 833–838. doi:10.1109/CICTN64563.2025.10932336.

Awards & National Examinations

2024	**UGC Junior Research Fellowship** *Computer Science and Applications* 99.95 percentile.
2025, 2022	**Graduate Aptitude Test in Engineering** *Computer Science and Information Technology* Qualified GATE CS in 2025 and 2022, and GATE Data Science and Artificial Intelligence in 2025.
2025	**Academic Merit Distinction** *MANIT Bhopal* Received a Merit Certificate at the 22nd Convocation for outstanding academic performance in the M.Tech. program.

Professional Experience

2021–2023 **Amdocs Development Center**
Associate Software Engineer, Gurugram

- Developed Python-based data-synchronization tools that reduced associated manual workload by approximately 90%.
- Designed database comparison and merging workflows to improve consistency across systems.

2020 **Stackfusion**
Machine Learning Intern, Gurugram

- Fine-tuned MobileNetSSD, ResNet, and YOLOv3 models for vehicle axle and licence-plate recognition.
- Developed Python-based preprocessing pipelines to improve training-data quality and diversity.

Patent

2025 **A Device for Producing Electricity Using Roof Air Ventilator**
Granted Design Patent, IP India
Design No. 440258-001, Class 13-01. Co-inventors: Anand Kumar, Ayush Kumar Agrawal, Uday Deo, Udit Deo, Shashank Kumar Soni, Bikramaditya Chakraborty, and Basundhara Singhdeo.

Technical Profile

Research Areas	Reliable Computer Vision; Adaptive Computation; Distribution Shift; Model Calibration; Vision Transformers; Semantic Segmentation; Efficient Deep Learning.
Programming	Python; C++; SQL; Shell scripting.
Frameworks	PyTorch; TensorFlow; OpenCV; Scikit-learn; NumPy; Pandas; MATLAB.
Methods	Experimental evaluation; uncertainty analysis; feature-space analysis; statistical modelling; multi-objective optimization.
Tools	Linux; Git; LaTeX; Flask.

Last updated: July 27, 2026